

3.51-inch LCD Quick Start

Keep your existing Raspberry Pi OS, applications, network settings, and user files. Choose the row that matches your system and Internet access.

ENGINEERING PREVIEW: Validate this package on test hardware before customer release.
 Supported target: Raspberry Pi Zero / Zero W / Zero 2 W with current Raspberry Pi OS overlays.

Before you start

1 Power off before attaching the LCD.	2 Align the 40-pin header carefully.
3 Remove other GPIO HATs for the first test.	4 Disconnect HDMI and use a stable 5V supply.

Choose the fastest method

System	Internet	Use	Fastest action
Raspberry Pi OS Desktop	Yes	Method 1	Open Terminal or connect by SSH, then paste the online command.
Raspberry Pi OS Desktop	No	Method 2	Transfer and extract the ZIP, then double-click INSTALL .
Raspberry Pi OS Lite / headless	Yes	Method 1	Connect by SSH and paste the online command.
Raspberry Pi OS Lite / headless	No	Method 2	Transfer and extract the ZIP, then run bash INSTALL .

METHOD 1 Online install - recommended

Use when the Raspberry Pi can reach the Internet. Open Terminal on the Pi or connect by SSH.

ONE COMMAND - paste the entire line below before pressing Enter.

```
curl -fLo /tmp/i https://github.com/iUniker/P035260107-easy-setup/raw/main/install.sh && sudo bash /tmp/i --reboot
```

The command downloads the complete script before running it. If your PDF app changes the line when copying, copy the online command from the GitHub README instead.

METHOD 2 Offline ZIP - no Internet required on the Pi

- Download the small offline ZIP** on any connected device.
- Transfer it to the Raspberry Pi and extract the complete ZIP.
- Desktop: double-click **INSTALL** and choose **Execute** or **Run** if prompted.

Raspberry Pi OS Lite / headless fallback:

```
bash INSTALL
```

After the automatic reboot

Allow 30-60 seconds for startup. Backlight alone is not a pass; the expected result is a readable picture.

Raspberry Pi OS Desktop	Raspberry Pi OS Lite
After 30-60 seconds, the LCD should show the desktop. Touch should act like a pointer and requires light pressure.	The LCD should show a text login console. A text screen is normal because Lite does not include a graphical desktop.

Troubleshooting

Symptom	Check first
White screen / backlight only	Confirm installation completed, disconnect HDMI, then check 40-pin alignment.
Completely dark	Check the 5V supply, USB cable, header alignment, and GPIO18 conflicts.
Picture works, touch does not	Remove other SPI devices; check GPIO27 and the touchscreen connector.
Wrong touch direction	Return the display to default orientation. Rotation also needs a touch transform.
Flicker / incorrect colours	Check power quality, header solder joints, connector seating, and panel batch.
Window does not fit	The native resolution is 480x320; some applications require a larger display.

Compatibility and conflicts

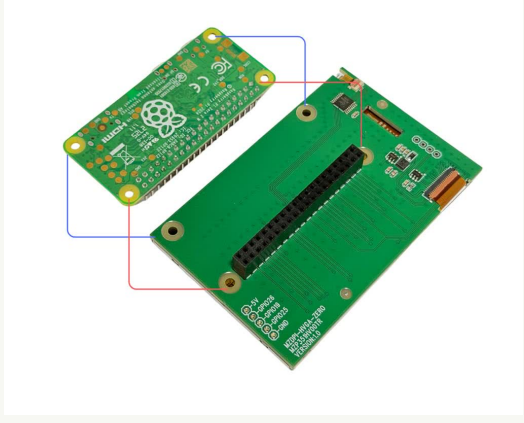
Supported: Raspberry Pi Zero, Zero W/WH, and Zero 2 W/2 WH with tested Raspberry Pi OS releases. Not yet supported: Ubuntu, RetroPie, Batocera, Kali, LibreELEC, Android, and custom kernels unless the exact release is listed as tested. GPIO conflicts: Other DPI displays, SPI0 devices, GPIO HATs, GPIO18, and GPIO27 devices may prevent the LCD or touch from working.

Need support?	Updates and downloads
From the extracted offline package, run: <pre>sudo bash diagnose.sh tee mzp351-diagnostic.txt</pre> Send the report with the Pi model, OS version, screen symptom, and a clear photo of the 40-pin connection.	github.com/iUniker/P035260107-easy-setup The diagnostic report does not collect passwords, Wi-Fi credentials, or user files.

Resistive touch requires light pressure from a fingertip, fingernail, or stylus. Multi-touch gestures are not supported.

Hardware & GPIO Reference

Connect the display only while power is disconnected. Use the mounting holes to confirm orientation before seating the 40-pin header.



Safe connection

- 1** Disconnect power and all cables.
- 2** Match all four mounting holes as shown.
- 3** Align every contact on the 40-pin header.
- 4** Press straight down and seat the header evenly.

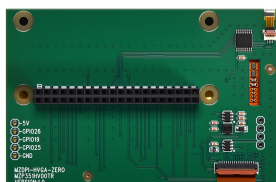
Never power the Pi with the connector offset by one row or one column. A misaligned connection can short the board.

GPIO use at a glance

The display uses 25 GPIOs. Physical pin numbers are shown in the two colored center columns.

Timing	Blue data	Green data	Red data	
Touch	Backlight	Not used	Power / GND	
	3.3V - power	1	2	5V - power
	GPIO2 - VSYNC	3	4	5V - power
	GPIO3 - HSYNC	5	6	GND
	GPIO4 - BLUE0	7	8	GPIO14 - GREEN2
	GND	9	10	GPIO15 - GREEN3
	GPIO17 - GREEN5	11	12	GPIO18 - BACKLIGHT
	GPIO27 - TOUCH IRQ	13	14	GND
	GPIO22 - RED2	15	16	GPIO23 - RED3
	3.3V - power	17	18	GPIO24 - RED4
	GPIO10 - TOUCH MOSI	19	20	GND
	GPIO9 - TOUCH MISO	21	22	GPIO25 - NOT USED
	GPIO11 - TOUCH SCLK	23	24	GPIO8 - BLUE4
	GND	25	26	GPIO7 - BLUE3
	GPIO0 - PCLK	27	28	GPIO1 - DATA ENABLE
	GPIO5 - BLUE1	29	30	GND
	GPIO6 - BLUE2	31	32	GPIO12 - GREEN0
	GPIO13 - GREEN1	33	34	GND
	GPIO19 - NOT USED	35	36	GPIO16 - GREEN4
	GPIO26 - NOT USED	37	38	GPIO20 - RED0
	GND	39	40	GPIO21 - RED1

On-board breakout



The LCD PCB exposes **5V**, **GPIO26**, **GPIO19**, **GPIO25**, and **GND**. GPIO19, GPIO25, and GPIO26 are not used by this display configuration.

The display still occupies the full 40-pin connector. Remove other HATs and SPI0 devices for the first test.

3.51-inch / 480x320 / RGB565 / 4-wire resistive touch / Raspberry Pi Zero family